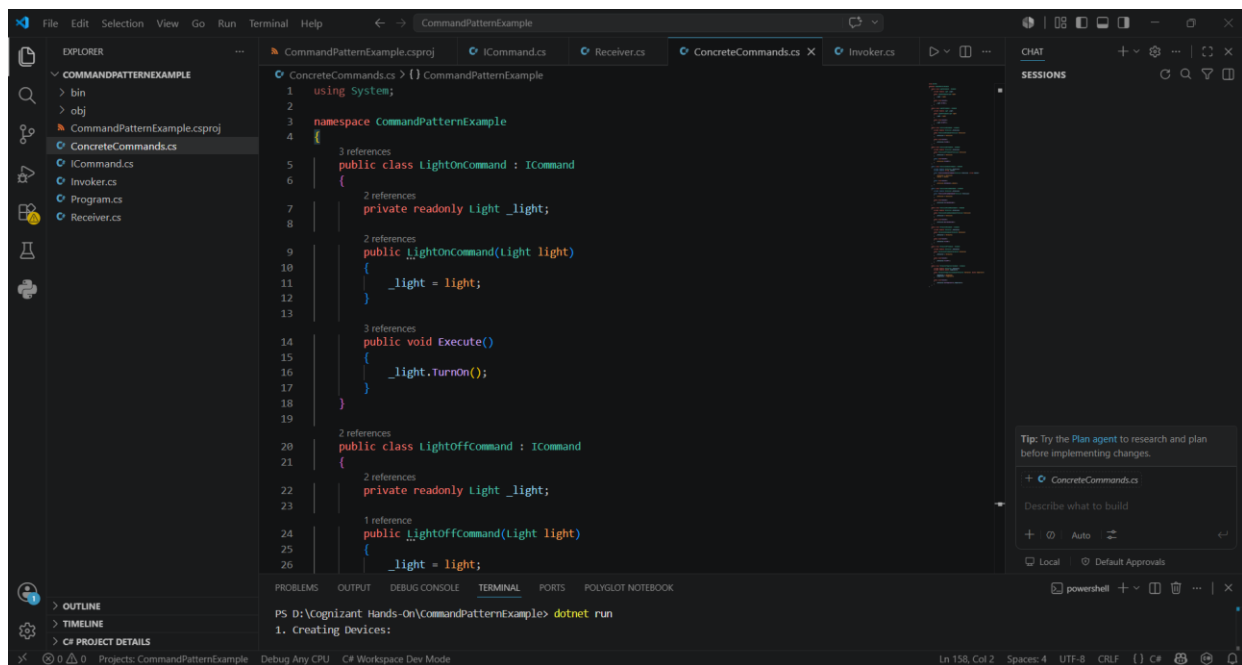


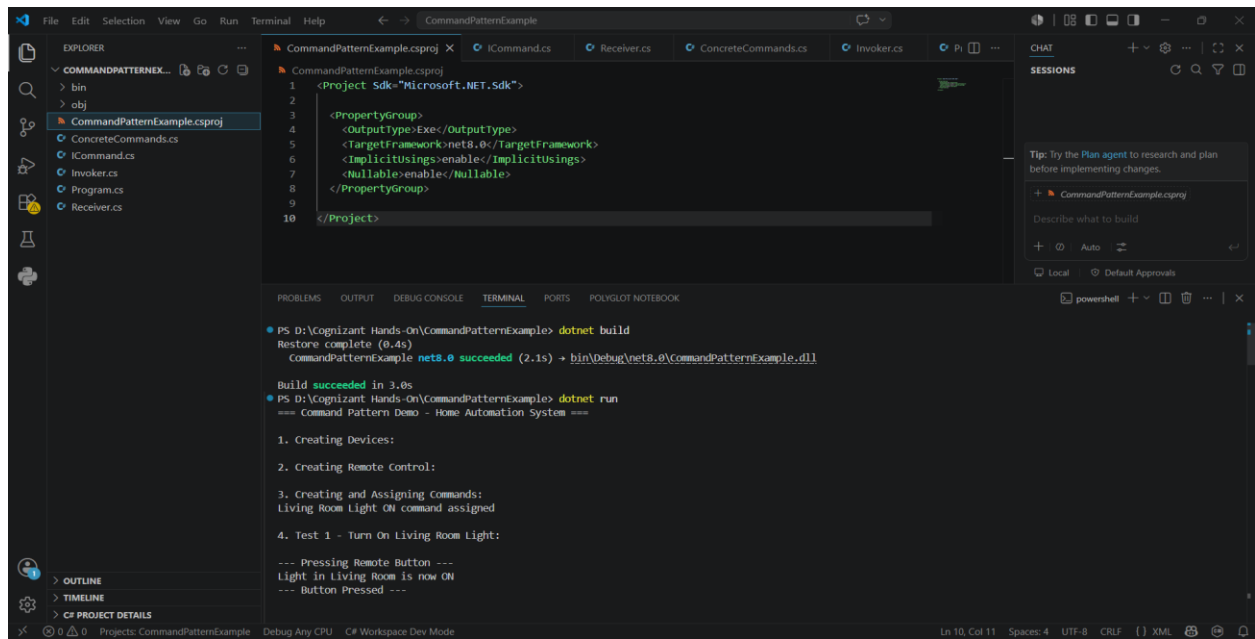
Hands-On

Exercise 9: Implementing the Command Pattern

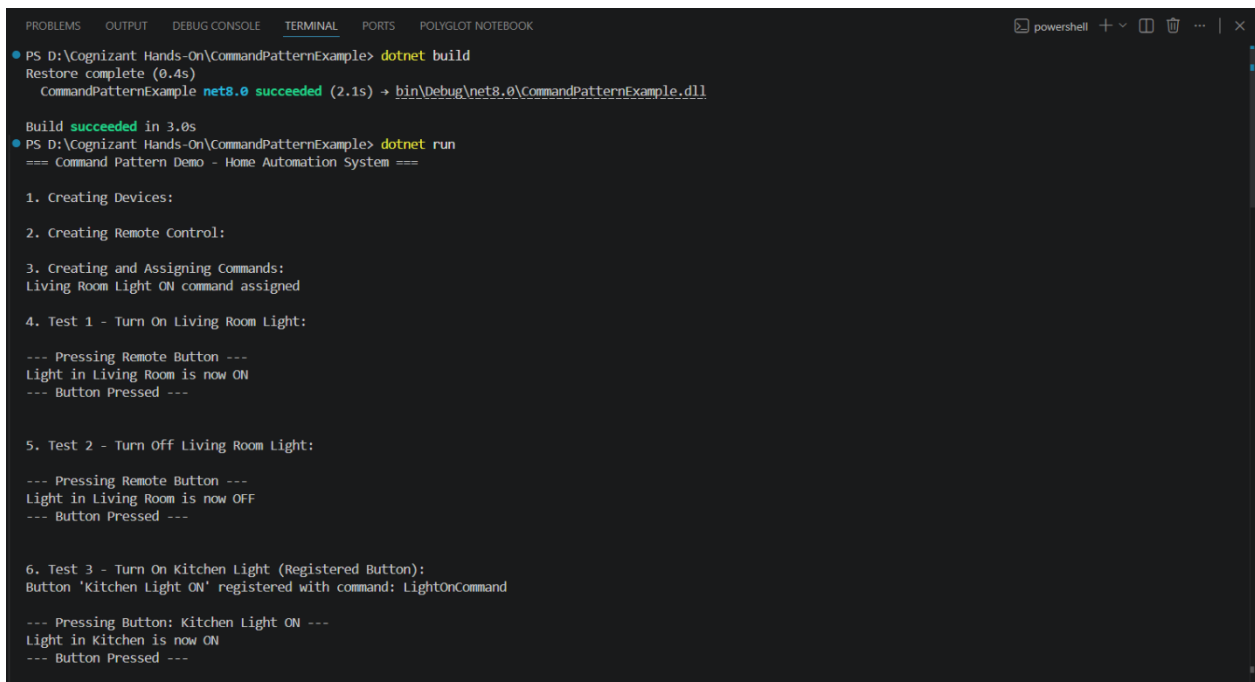
I implemented, executed and tested the Command Pattern in .NET. I have created a project called CommandPatternExample. I have developed a Command interface with Execute() method and concrete command classes like LightOnCommand and LightOffCommand to wrap requests for the control of a light device. I also created a Light class that was the receiver and had methods to switch the light on and off, and a RemoteControl class that was the invoker, holding and executing command objects. Once it was implemented, I executed the program and sent various commands using the RemoteControl to check that the correct actions were taken on the Light object. This practical exercise gave me a good insight into how the Command Pattern can be used to represent requests as objects, how it can be used to separate the sender from the receiver, and how it can be used to implement queuing, logging, and undoable operations in .NET applications.

Coding:





Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK powershell + v [ ] [ ] ... | X

PS D:\Cognizant Hands-On\CommandPatternExample> dotnet run
7. Test 4 - Turn On Bedroom TV:
Button 'TV ON' registered with command: TelevisionOnCommand

--- Pressing Button: TV ON ---
Television in Bedroom is now ON
Volume: 10, Channel: News
--- Button Pressed ---

8. Test 5 - Change TV Channel:
Button 'TV Channel Sports' registered with command: TelevisionChannelCommand

--- Pressing Button: TV Channel Sports ---
Television channel changed to: Sports
--- Button Pressed ---

9. Test 6 - Increase TV Volume:
Button 'TV Volume Up' registered with command: TelevisionVolumeUpCommand

--- Pressing Button: TV Volume Up ---
Television volume increased to: 11
--- Button Pressed ---

--- Pressing Button: TV Volume Up ---
Television volume increased to: 12
--- Button Pressed ---

10. Test 7 - Turn On Office Thermostat:
Button 'Thermostat ON' registered with command: ThermostatOnCommand

--- Pressing Button: Thermostat ON ---
Thermostat in Office is now ON
Current Temperature: 70F
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK powershell + v [ ] [ ] ... | X

PS D:\Cognizant Hands-On\CommandPatternExample> dotnet run
11. Test 8 - Set Thermostat Temperature:
Button 'Set Temp 75F' registered with command: ThermostatTemperatureCommand

--- Pressing Button: Set Temp 75F ---
Thermostat temperature set to: 75F
--- Button Pressed ---

12. Test 9 - Show Command History:

=== Command History ===
Button: Kitchen Light ON -> Command: LightOnCommand
Button: TV ON -> Command: TelevisionOnCommand
Button: TV Channel Sports -> Command: TelevisionChannelCommand
Button: TV Volume Up -> Command: TelevisionVolumeUpCommand
Button: Thermostat ON -> Command: ThermostatOnCommand
Button: Set Temp 75F -> Command: ThermostatTemperatureCommand
=====

13. Test 10 - Undo Functionality:
Commands in undo stack: 2

--- Pressing Undo Button ---
Undoing last command: LightOffCommand
--- Undo Complete ---

14. Test 11 - Multiple Devices:

--- Pressing Remote Button ---
Light in Living Room is now ON
--- Button Pressed ---

--- Pressing Remote Button ---
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  POLYGLOT NOTEBOOK
PS D:\Cognizant Hands-On\CommandPatternExample> dotnet run

--- Pressing Remote Button ---
Television in Bedroom is now ON
  Volume: 10, Channel: News
--- Button Pressed ---

--- Pressing Remote Button ---
Thermostat in Office is now ON
  Current Temperature: 75F
--- Button Pressed ---

15. Test 12 - Turn Off All Devices:

--- Pressing Remote Button ---
Television in Bedroom is now OFF
--- Button Pressed ---

--- Pressing Remote Button ---
Thermostat in Office is now OFF
--- Button Pressed ---

=== Command Pattern Demo Complete ===
Key Benefits:
  Encapsulates Requests: Commands encapsulate action and receiver
  Decouples Invoker and Receiver: Remote doesn't know about devices
  Easy to Extend: Add new commands without modifying existing code
  Supports Undo: Command history enables undo operations
  Macro Commands: Can combine multiple commands into one
PS D:\Cognizant Hands-On\CommandPatternExample> 
```